

MTH 203: Sequences and series

Mid-examination 2: 11/10/2025,

Time: 12:00 to 13:00

Total points: 25

Problem 1. (1 point each) State whether the following statements are **true** or **false**. (No proofs required.)

- (a) Every subsequence of an unbounded sequence is divergent.
- (b) Suppose $(a_n)_{n \in \mathbb{N}}$ and $(b_n)_{n \in \mathbb{N}}$ are sequences such that $\lim_{n \rightarrow \infty} a_n = 1$ and $(b_n)_{n \in \mathbb{N}}$ is divergent. Let $(c_n)_{n \in \mathbb{N}}$ be the sequence defined by $c_n = a_n b_n$ for all n . Then $(c_n)_{n \in \mathbb{N}}$ is a divergent sequence.
- (c) The series $\sum_{n=1}^{\infty} x^n$ converges if and only if the sequence $(x^n)_{n \in \mathbb{N}}$ is convergent.
- (d) If $(a_n)_{n \in \mathbb{N}}$ is a bounded sequence such that $a_n \geq 0$ for all n , then it is necessarily a convergent sequence.

Problem 2. (3 points each) Calculate the following:

(a)

$$\lim_{n \rightarrow \infty} \left(\frac{\sqrt{n} - 1}{\sqrt{n} + 1} \right)$$

(b)

$$\sum_{n=1}^{\infty} \frac{2^{3n}}{3^{2n}}$$

You must justify all your calculations.

Problem 3. (5 points) Let $(a_n)_{n \in \mathbb{N}}$ be a sequence such that $\lim_{n \rightarrow \infty} a_n = 0$. Prove that there exists a subsequence $(a_{n_k})_{k \in \mathbb{N}}$ of this sequence such that the series $\sum_{k=1}^{\infty} a_{n_k}$ converges.

Problem 4. (5 points) Let $(x_n)_{n \in \mathbb{N}}$ be a sequence and let L be a real number. Suppose that every subsequence of $(x_n)_{n \in \mathbb{N}}$ has a subsequence which converges to L . Prove that $\lim_{n \in \mathbb{N}} x_n = L$. (Warning: Read the second sentence carefully.)

Problem 5. (5 points) Let $(x_n)_{n \in \mathbb{N}}$ be a sequence such that $x_n \geq 0$ for all n . Let $(y_n)_{n \in \mathbb{N}}$ be the sequence defined by $y_n = (-1)^n x_n$ for all n . If the sequence $(y_n)_{n \in \mathbb{N}}$ converges, prove that $(x_n)_{n \in \mathbb{N}}$ also converges.